the following 5 daughters. Calculate for a body weight of 70 kg the number of radioactive decays per unit time from ^{3}H, ^{14}C, ^{40}K and ^{226}Ra. Assume 30 TU in water.

15.2. Using the information above, how many grams of the body's molecules (assume average mole weight of 10^5) will be damaged in a year if the G(damage) value is 3.1×10^{-7} mol/J? Assume $E_{abs}(\beta) = E_{max}/3$.

15.3. Under the same assumption as above, what amount of damage will be caused by cosmic radiation? Assume that the cosmic particles produce 3×10^9 ion pairs s^{-1} m^{-3} of the body.

15.4. With the information in exercise 15.1, calculate annual doses received from (a) ^{40}K, and (b) ^{226}Ra and daughters. Assume $w_r(\alpha) = 20$.

15.5. Ten mg ^{238}U has been collected in the kidneys. Considering the biological half-life of uranium and assuming only one α-emission in ^{238}U decay, calculate the dose (in Sv) received by the organ if the uranium is evenly distributed. The weight of a kidney is 150 g.

15.6. A γ-dose rate of 1 Sv is assumed to inactivate (kill) human cells. The body contains 6×10^{13} cells in a cell weight of 42 kg for a 70 kg man. (a) What average energy (in eV) has to be deposited in a cell to kill it? (b) Calculate the number or kidney cells destroyed for the dose received in exercise 15.5. For simplicity assume the cells to be cubic with a side length of about 11 μm.

15.7. A tumor has the weight of 80 g and we wish to destroy 20% of the cells by irradiating with 180 MeV protons with such penetration that half of the energy is deposited in the tumor. The particle beam is 5 μA. For what time must the irradiation be? A cell of weight 10^{-9} g is assumed to be killed on the absorption of 200 keV and no cell is assumed to be killed twice.

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CHAPTER 16

Particle Accelerators

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This chapter describes the sources – accelerators – that can be used to obtain energetic charged particles, and secondarily neutrons and photons, that can induce nuclear reactions or chemical changes. Thus we include various accelerators for producing beams of charged particles as well as targets for neutron production. Electrons can be accelerated by the same principles as positive ions but in the following paragraphs we will focus on the acceleration of positive ions because these are more useful for nuclear reactions. Although accelerators can be used to produce neutrons, a more copious neutron source is the nuclear reactor, see Chapter 19. Small neutron sources, consisting of the spontaneous fissioning nuclide ^{252}Cf or of mixtures of elements like radium and beryllium in which nuclear transmutations (mostly (α,n)-reactions) produce neutrons, are discussed in §10.9.

16.1. CHARGED PARTICLE ACCELERATORS

In order to induce nuclear reactions with positively charged projectiles such as protons, deuterons, α-particles, oxygen ions, or uranium ions, it is necessary that the projectile particles have sufficient kinetic energy to overcome the Coulomb barrier created by the repulsion between the positive charges of the projectile and the target nucleus. While there is some probability that a positive projectile can tunnel through the Coulomb barrier at kinetic energies lower than the maximum value of the

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